

ECON1010

Introductory Microeconomics

Tutorial 8 | Imperfect Competition – Monopoly



This Week in ECON1010

CML 4 is due **next week** 4.00pm May 05

Public Holiday **next Monday**

ECON1010 Class Timetable

Semester 1 2026

Time	Mon	Tue	Wed	Thu	Fri
8:00-8:30			TUT 3 - NIK		
8:30-9:00					
9:00-9:30	TUT 4 - SAM	TUT 8 - BORIS	TUT 12 - NIK	TUT 7 - GABE	
9:30-10:00					
10:00-10:30	TUT 9 - SAM			TUT 20 - LUCAS	TUT 19 - DYLAN
10:30-11:00					TUT 15 - NANDAN
11:00-11:30		TUT 18 - EDWIN TUT 5 - GABE			TUT 2 - DYLAN
11:30-12:00					
12:00-12:30					TUT 13 - NANDAN
12:30-13:00					
13:00-13:30	TUT 21 - BORIS	TUT 6 - EDWIN	TUT 14 - IKMAL		LECTURE
13:30-14:00					
14:00-14:30					
14:30-15:00					
15:00-15:30		TUT 10 - IKMAL	TUT 16 - KIRTAN		TUT 11 - LUCAS TUT 17 - GUANCHU
15:30-16:00					
16:00-16:30					
16:30-17:00					
17:00-17:30			TUT 1 - KIRTAN		
17:30-18:00					
18:00-18:30					
18:30-19:00					

**Public Holiday
next week –
attend any
other tutorial!**

You can still come on Monday, but there will be no class, and I will not be here.

Assessment

85% **7**

75% **6**

65% **5**

50% **4**

47% **3**

30% **2**

0% **1**

CMLs (25%)

- Computer Managed Learning
- Each worth 5%
- Best 5/6 count towards final grade

CLEAR-JE (20%)

- Contextualised Learning of Economics via Authentic Reflection using Journal Entries
- Each worth 10%
- Best 2/3 count towards final grade

Final Exam (55%)

Tools and Resources

I need HELP...

1. **Ed Discussion Board**
(Blackboard)
2. **Consultation** (8.00am
Mon, 39-125A)
3. **Staff email**
econ1010@uq.edu.au

I REALLY need HELP...

- <https://my.uq.edu.au/contact/student-central>

ECON1010 Consult Timetable

Semester 1 2026

Time	Mon	Tue	Wed	Thu	Fri
8:00-8:30	CONSULT - SAM 39-125A				
8:30-9:00					
9:00-9:30					
9:30-10:00					
10:00-10:30				CONSULT - LUCAS 39-125A	
10:30-11:00					
11:00-11:30	CONSULT - GABE 39-125A	CONSULT - BORIS 39-125A	CONSULT - IKMAL 39-125A	CONSULT - NIK 39-125A	CONSULT - CARL 39-641
11:30-12:00					
12:00-12:30					
12:30-13:00					
13:00-13:30					LECTURE
13:30-14:00		CONSULT - DYLAN 39-125A			
14:00-14:30				CONSULT - GUANCHU 39-125A	
14:30-15:00					
15:00-15:30					
15:30-16:00					
16:00-16:30			CONSULT - KIRTAN 39-125A	CONSULT - NANDAN 39-125A	
16:30-17:00					
17:00-17:30					

Student Central

Need help? We've got you covered.

Contact us

Monday to Friday

- ✉ Email
- ☎ 1300 275 870 (8.30am-5pm)
- 📍 Building 42, St Lucia
- 📍 Building 8101A, Gatton

Chat – 1:51 (1 Minute) wait

Emergency help

For immediate risk:

- UQ Campus Security
- ☎ 07 3365 3333 (24/7)

Off-campus emergency

- ☎ 000 (24/7)

Crisis support

For urgent mental health support:

- UQ Counselling and Crisis Line
- ☎ 1300 851 998 (24/7)

Text a Crisis Counsellor

- ☎ 0488 884 115 (4.30pm-8am)

Sexual Misconduct Support Unit (SMSU)

Imperfect Competition – Monopoly

Market Structures

Increasing Competition



**Perfect
Competition**

**Monopolistic
Competition**

Oligopoly

Monopoly

Increasing Concentration



Question 1a

In your own words:

- i) What are characteristics of a monopoly?
- ii) What can cause a monopoly to exist?
- iii) What quantity should a monopoly produce to maximise profit?
- iv) Why would a firm with market power not be able to sell any quantity it wishes at any price?



Question 1a

i) What are characteristics of a monopoly?

- **Only seller of a product/service**
- **No close substitutes**
- **Firm faces downward sloping demand curve**
- **Firm is a price maker – flexibility to choose a price (but can only maximise profit at a specific output)**

ii) What can cause a monopoly to exist?

- **Entry into industry deliberately blocked: patents, copyright, government permission**
- **Control over key raw material resources**
- **Network externalities: products becoming more useful when more consumers adopt it**
- **Economies of scale: some firms can operate at lower marginal costs when they own more of the market. Other firms cannot compete with their low prices and large market share.**

Question 1a

iii) What quantity should a monopoly produce to maximise profit?

In this course, it is sufficient to know and be able to apply the rule $MR = MC$. This will help you determine the profit maximising output for a monopoly. For those that want to know more, you can refer to the lecture notes “proof” of $MR = MC$.

iv) Why would a firm with market power not be able to sell any quantity it wishes at any price?

As a firm with market power is a price maker, it can set its own price. The firm, however, will only be able to sell the quantity of goods that corresponds to their demand curve. As such, the firm cannot freely choose how much quantity to sell at any given price.

Question 1b

Consider the following table. It costs a firm \$6 to produce each additional unit of output. How many units of output will a profit maximising monopolist produce and what price will they set?

Quantity (units)	Price (\$/unit)	Total Revenue (\$)	Marginal Revenue (\$/unit)	Marginal Cost (\$/unit)
0	11			
1	10			
2	9			
3	8			
4	7			
5	6			



Question 1b

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0	11	0		
1	10	10		
2	9	18		
3	8	24		
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5	6			

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0	11	0	-	
1	10	10		
2	9	18		
3	8	24		
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3	8	24	6	
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4	7	28	4	6
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3	8	24	6	6
4	7	28	4	6
5	6	30	2	6

The profit maximising quantity is found where $MR = MC$.
 $MR (\$6/\text{unit}) = MC (\$6/\text{unit})$ at the output quantity of 3 units.
The price will be \$8/unit

Question 2a

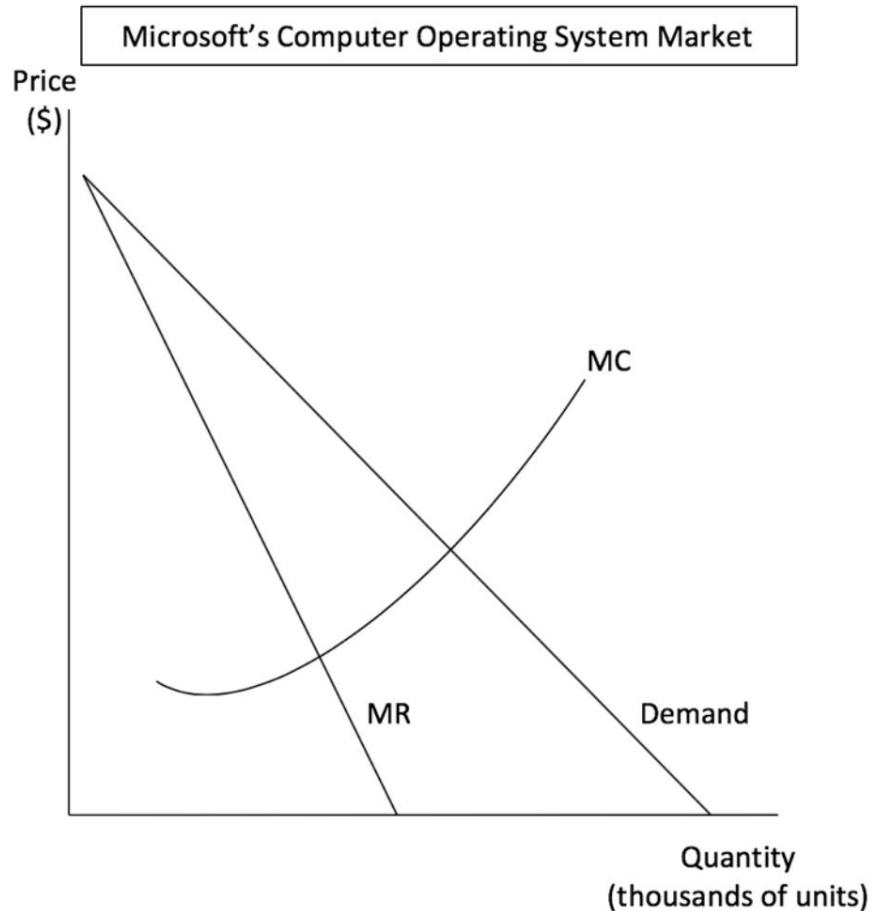
Microsoft can be considered to have a monopoly in the computer operating system market.

- i) Draw a hypothetical demand curve, marginal revenue curve, and marginal cost curve to represent the dominance of Microsoft software in the market.
- ii) Show the profit maximising output quantity and price Microsoft will strive for.
- iii) In your diagram, add an average total cost curve for Microsoft so as to show it making an economic profit.
- iv) Does Microsoft's dominance in the software market impose a loss on society in any way compared to a competitive market outcome? Refer to your diagram in iii) to answer this question and interpret this answer by referring to what physically takes place in the market.



Question 2a

i) Draw a hypothetical demand curve, marginal revenue curve, and marginal cost curve to represent the dominance of Microsoft software in the market.

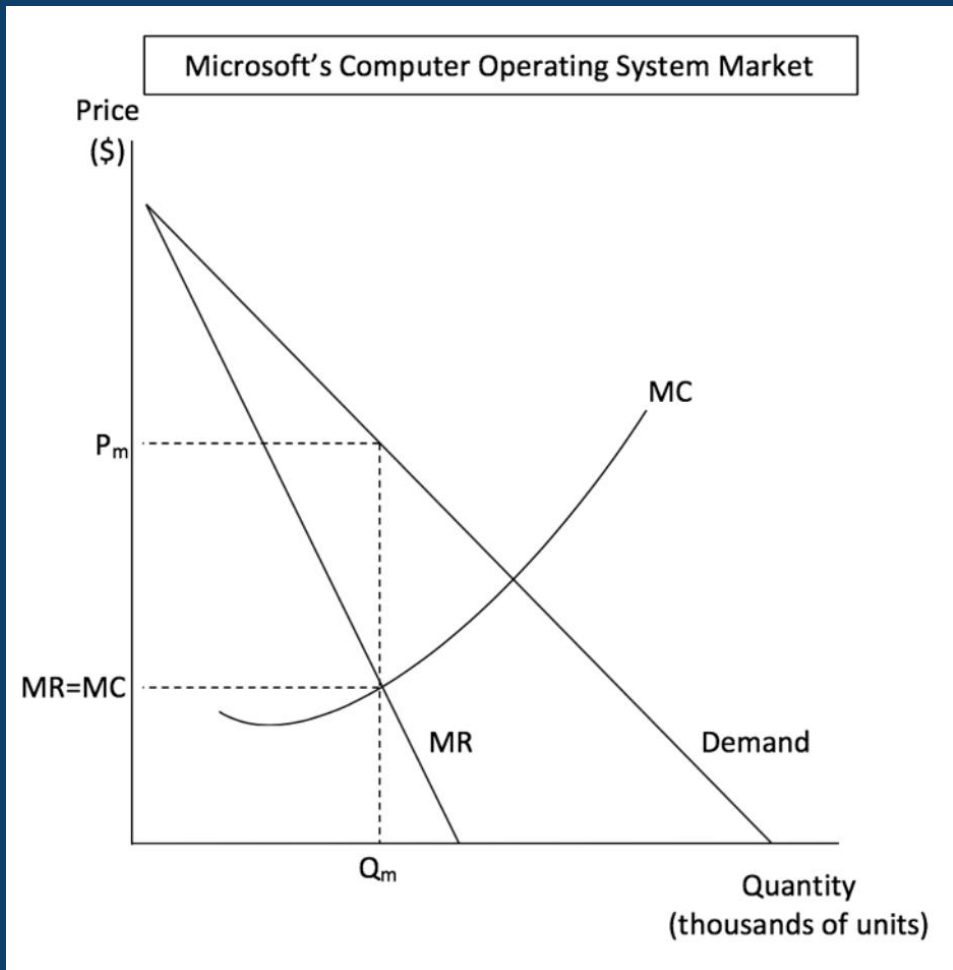


Sam's Steps

1. Draw downward sloping demand curve
2. Draw MR curve, which is always half of (twice as steep as) demand curve
3. Draw MC curve (NIKE Swoosh)

Question 2a

ii) Show the profit maximising output quantity and price Microsoft will strive for.

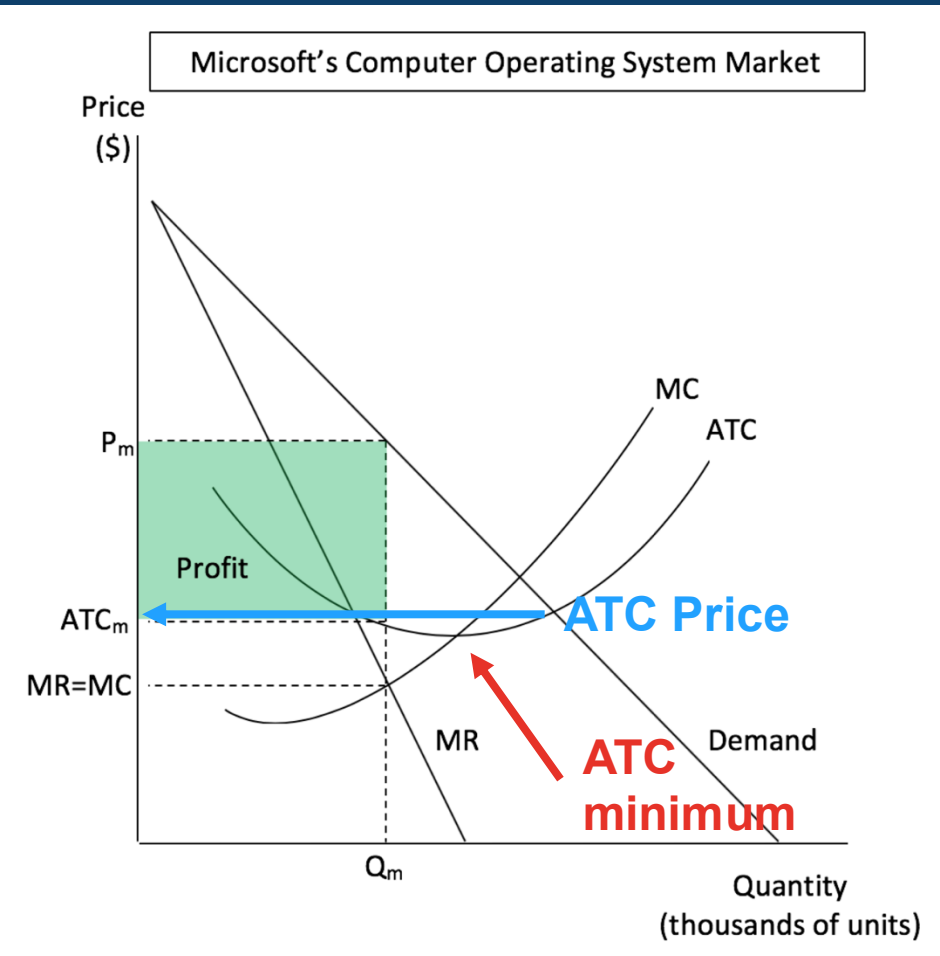


Sam's Steps

1. Draw downward sloping demand curve
2. Draw upward sloping marginal cost curve
3. Draw steeper downward sloping marginal revenue curve
4. Revenue to profit maximise, they should sell Q_m
5. Label units of software and set a price at P_m .
6. Trace the profit max price (P_m)

Question 2a

iii) In your diagram, add an average total cost curve for Microsoft so as to show it making an economic profit.



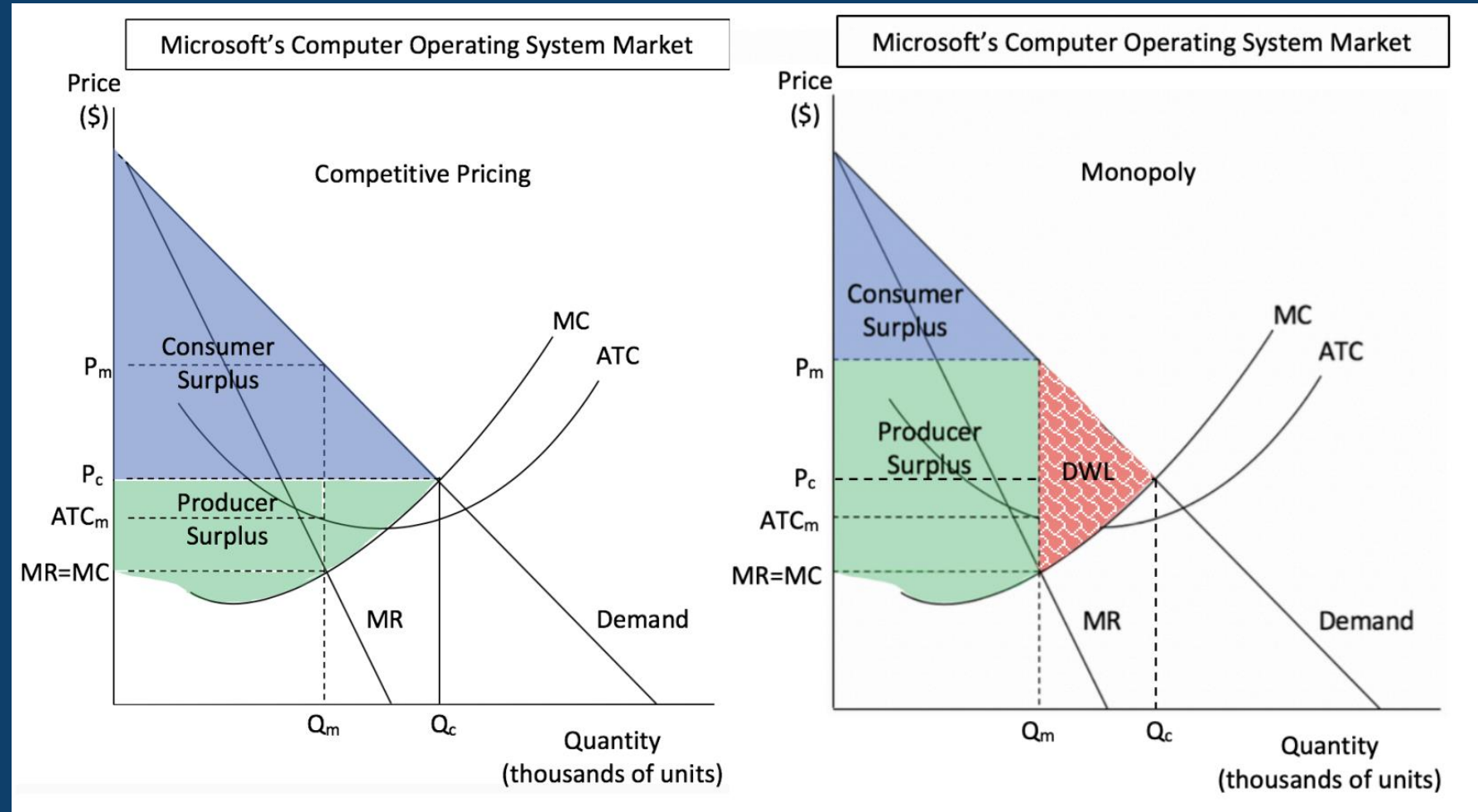
Sam's Steps

1. Draw downward sloping demand curve
2. Draw MR curve, which is always half of (twice as steep as) demand curve
3. Draw MC curve (NIKE Swoosh)
4. Recall profit max is where $MC = MR$
5. Label Q^* as profit max quantity (Q_m)
6. Trace Q^* up to demand curve, and label P^* as profit max price (P_m)
7. For economic profit, ATC must be less than price at profit max quantity
8. MC will intersect ATC at **ATC minimum** ALWAYS
9. Label **price at intersection of ATC and profit max q** (ATC_m)
10. Difference between ATC price and P^* is **economic profit** for each unit sold

Question 2a

iv) Does Microsoft's dominance in the software market impose a loss on society in any way compared to a competitive market outcome? Refer to your diagram in iii) to answer this question and interpret this answer by referring to what physically takes place in the market.

As Microsoft has patents on their software, they operate as a monopoly. This allows them to set their software price to maximise profit. Microsoft will set prices higher than they otherwise would in a competitive market to receive higher producer surplus (at the expense of the consumer surplus). The diagrams below show the loss to society in the hatched area (DWL).



Question 2b

Wet'n'Wild on the Gold Coast is considered Australia's greatest water park. Due to the water park's huge range of rides and popular reputation it is effectively a monopoly in the Australian water park market. **Each visitor entry into the water park costs Wet'n'Wild an additional \$40/entry** in staff, lifeguards, electricity and opportunity costs. Below is a table of potential prices Wet'n'Wild could charge per entry and the corresponding number of entries per day.

As Wet'n'Wild can exert monopoly power, they set a price of \$160 per entry without consulting any economic concepts.

- i) Expand the price table above by adding additional columns so as to include the Total Revenue, Marginal Revenue and Marginal Costs of Wet'n'Wild.
- ii) What is Wet'n'Wild's marginal revenue when the price is \$160 per entry?
- iii) At \$160 per entry, should Wet'n'Wild decrease or increase the number of entries to maximise profit?
- iv) What price per entry should Wet'n'Wild be charging in order to maximise profit?

Price (\$/entry)	Quantity (entries/day)
200	4,000
160	6,000
120	10,000
80	20,000
40	60,000



Question 2b

i) Expand the price table above by adding additional columns so as to include the Total Revenue, Marginal Revenue and Marginal Costs of Wet'n'Wild.

Price (\$/entry)	Quantity (entries/day)	Total Revenue (\$)	Marginal Revenue (\$/unit)	Marginal Cost (\$/unit)
200	4,000	800,000	-	-
160	6,000	960,000	80	40
120	10,000	1,200,000	60	40
80	20,000	1,600,000	40	40
40	60,000	2,400,000	20	40

Important reminder:

$$MR = \frac{\Delta TR}{\Delta Q}$$

Question 2b

ii) What is Wet'n'Wild's marginal revenue when the price is \$160 per entry?

Price (\$/entry)	Quantity (entries/day)	Total Revenue (\$)	Marginal Revenue (\$/unit)	Marginal Cost (\$/unit)
200	4,000	800,000	-	-
160	6,000	960,000	80	40
120	10,000	1,200,000	60	40
80	20,000	1,600,000	40	40
40	60,000	2,400,000	20	40

Marginal Revenue when the prices is \$160/entry is \$80/entry.

iii) At \$160 per entry, should Wet'n'Wild decrease or increase the number of entries to maximise profit?

Wet'n'Wild should increase the number of entries on the basis that the MR is greater than the MC at \$160 per entry in order to maximise profits.

Question 2b

Price (\$/entry)	Quantity (entries/day)	Total Revenue (\$)	Marginal Revenue (\$/unit)	Marginal Cost (\$/unit)
200	4,000	800,000	-	-
160	6,000	960,000	80	40
120	10,000	1,200,000	60	40
80	20,000	1,600,000	40	40
40	60,000	2,400,000	20	40

iv) What price per entry should Wet'n'Wild be charging in order to maximise profit?

Wet'n'Wild should decrease prices to \$80/entry and increase quantity to 20,000 visitors per day to maximise their profit (to reach the situation where $MR = MC = \$40/\text{entry}$).

Question 3

Work through the following to develop your understanding of how monopolies affect your consumer surplus.

- i) Think of a popular company that is recognised as having a monopoly.
- ii) How has this company been able to gain monopoly power?
- iii) How could the government intervene to remove the source of the company's monopoly power?
- iv) How would this hypothetical intervention affect you?



Thank You

Next week: Imperfect Competition – Strategy

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